RAJALAKSHMI ENGINEERING COLLEGE

Lavanya V 250701386-2025-CSE

L2

Started on	Friday, 25 September 2026, 2:12 PM
State	Finished
Completed on	Friday, 25 September 2026, 2:14 PM
Time taken	1 min 49 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*
Output:*6*
Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1
3+3
1+1+1+3
1+1+3+1
1+3+1+1
3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    if (n == 0 || n == 1 || n == 2) {
        printf("1");
        return 0;
    }
    if (n == 3) {
        printf("2");
        return 0;
    }

    long long a = 1, b = 1, c = 2, d;

    for (int i = 4; i <= n; i++) {
        d = c + a;
        a = b;
        b = c;
        c = d;
    }

    printf("%lld", c);
    return 0;
}
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Back to Course